MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPES 3RP1 AND 3RP1A

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Electrostatic deflection and focus (3RP1 has a curved face and the 3RP1A has a flat face)

PIN CONNECTIONS AND DIMENSIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	ed	Eb1	Eb2	Light output	Rg	Zd	Alt
Unit:	V	Vdc	v	Vdc	Vdc	ftL	Meg	Meg	ft
Maximum:	6.9	0, -200	550	1,100	2,750	---	1.5	1.0	10,000
Minimum:	5.7	---	---	---	500 (see note 1)	4.0	---	---	---
TEST CONDITIONS:	6.3	Adjust	---	Focus	1,000	---	---	---	---

GENERAL:

Qualification - Required

(D) denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
				MIN	MAX	
	Qualification inspection					
1216	Base material insulating quality	Zone 5 (minimum)	---	---	---	---
1141	Pressure (implosion)		---	---	---	---
5111	Vibration		Width	---	1.0	mm
1331	Direct interelectrode capacitance	g1 to all	Cg1	---	11.0	pF
		D1 to D2	C1D2	---	4.0	pF
		D3 to D4	C3D4	---	4.0	pF
		D1 to all except D2	CD1	---	15.0	pF
		D2 to all except D1	CD2	---	11.0	pF
		D3 to all except D4	CD3	---	10.0	pF
		D4 to all except D3	CD4	---	11.0	pF
5101	Neck and bulb alinement (electrostatic types)		Dia	---	1.63	in.
5248	Deflection-factor uniformity		---	---	---	---
	Quality conformance inspection, part 1					
5106	Bulb, screen, and face-plate quality		---	---	---	---
5201	Voltage breakdown		---	---	---	---
5201	Voltage breakdown (electrostatic types)		---	---	---	---
5206	Gas "cross"	Light = 4 ftL (see note 2)	---	---	---	---
5221	Light output	See note 2	Light	4.0	---	ftL
5223	Modulation	Light = 4 ftL	ΔEc1	---	48	Vdc
5231	Spot position (electrostatic deflection)		---	---	15.0	mm
5231	Spot displacement (leakage)		Displ	---	7.0	mm
5241	Grid cutoff voltage		Ec1	---	-67.5	Vdc
5251	Grid No. 1 leakage current		---	---	---	---
5251	Anode No. 1 leakage current		---	---	---	---
5251	Anode No. 2 leakage current		---	---	---	---
	Quality conformance inspection, part 2					
1101	Secureness of base, cap or insert		---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued					
(D) 1111	Base pin solder depth		---	---	---	---
1301	Heater current		If	540	660	mA
5201	Electrode current (anode No. 1)	Ec1 = 0	Ib1	-15	10	μAdc
5201	Electrode current (cathode)	Light = 4 ftL	Ik	---	1,000	μAdc
5101	Neck and base alinement (electrostatic types)		---	---	---	---
5101	Base and trace alinement	3D4; pin No. 1	---	---	---	---
5101	Angle between traces		---	---	---	---
5216	Stray light emission (conventional types)	Eb2 2,750 Vdc	---	---	---	---
5226	Line width A (electrostatic deflection)	Light = 4 ftL	Width	---	0.65	mm
5226	Line width B (electrostatic deflection)	Light = 4 ftL	Width	---	0.65	mm
5246	Focusing voltage at cutoff	Light 4 ftL	Eb1	165	310	Vdc
5246	Focusing voltage at modulation condition		Eb1	165	310	Vdc
5248	Deflection factor	1D2	DF	73	99	Vdc/in.
5248	Deflection factor	3D4	DF	52	70	Vdc/in.
5251	Heater-cathode leakage current		---	---	---	---
1105	Permanence of marking		---	---	---	---
	Quality conformance inspection, part 3					
---	Life test	Group C, light = 4 ftL; Eb2 = 2,750 Vdc, t = 500 hours (min)	---	---	---	---
---	Life-test end points:	Light = 3.4 ftL	---	---	---	---
	Line width A		Width	---	0.65	mm
	Line width B		Width	---	0.65	mm
	Modulation		ΔEc1	---	48	Vdc
	Grid No. 1 leakage current		---	---	---	---
	Anode No. 1 leakage current		---	---	---	---
	Anode No. 2 leakage current		---	---	---	---
	Heater-cathode leakage current		---	---	---	---
	Stray light emission		---	---	---	---

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NOTES:

1. This value is recommended only for low velocity deflecting and low ambient light levels.
2. This test to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11, 17
DSA - ES
Army - MI, MU

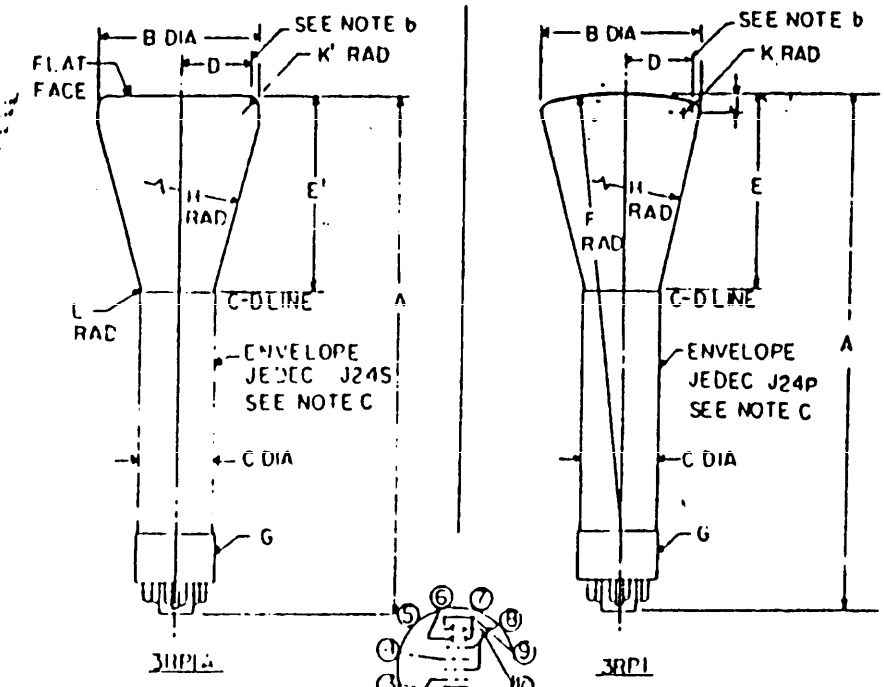
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH

Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2532)

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Ltr	Dimensions are in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
G	Base: B12-43 or B10-75	
Quality conformance inspection, part 2		
A	8.875(225.43)	9.375(238.13)
B	2.938 (74.63)	3.063 (77.80)
C		1.438 (36.53)
D	1.380 (35.05)	
Reference dimensions (see note a)		
E	3.875 (98.43)	
E'	3.563 (90.50)	
F	8.000(203.20)	
H	17.000(431.80)	
J	0.350 (8.89)	
K	0.250 (6.35)	
K'	0.125 (3.18)	
L	0.500 (12.70)	

NOTES:
a. These dimensions are for information only and are not required for inspection purposes.
b. Useful screen radius.
c. Clear or filter glass.

FIGURE 1. Outline drawing of electron tube types 3RP1 and 3RP1A.

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-[illegible]/2000 [illegible], RADIOFRAT TYPES 3CP1 AND 3CP1A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-PM 1068-68

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